

Harvest index at maize in different growing conditions

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Abstract

The harvest index represents the proportion of plant dry biomass allocated into grains. In order to explore the high potential of maize plants to produce grain yields it is necessary to know and even more important to understand how the harvest index is influenced by the different environmental factors and management practices. Climatic and soil conditions are two of the most important factors affecting harvest index. Among the agronomic practices with a great influence upon the harvest index there are counted the sowing patterns, respectively row spacing and plant density, but also preceding crop and soil tillage. The objective of this paper is to present the results obtained at different hybrids of maize regarding the harvest index under the specific growing conditions from South Romania in the period 2012-2014 and under different sowing conditions, respectively different row spacing, plant densities, preceding crops, and soil tillage. In our field experiments, the average values of the harvest index at maize varied between 0.20 to 0.56, more frequently between 0.3 and 0.5. Harvest index registered higher values under more favourable soil and climatic conditions than under less favourable conditions. Narrow rows, especially the row spacing of 50 cm, tended to determine smaller variation of the harvest index according to plant density and maize hybrid. The highest average values of the harvest index were registered at plant density of 70.000 plants ha⁻¹. Maize as preceding crop determined higher values of the harvest index than sunflower, and disking determined higher values of the harvest index than plowing.

Key words: *harvest index, maize hybrid, growing conditions, plant density, row spacing, preceding crop, soil tillage*

1. Introduction

Maize (*Zea mays* L.) is a cereal crop belonging to the *Poaceae* (*Gramineae*) family, which is ranking the third place among the most important crops based on harvested area and production.

As harvested area of grain maize, Romania ranges the first place in European Union. In Romania, the grain maize is the most important crop after harvested area, which varied in the last decade between 2 and 3.2 million hectares. The high surfaces cultivated with maize are explained by the fact that Romania has very favourable growing conditions for maize.

The harvest index represents the proportion of plant biomass allocated into grains; it describes the plant capacity to allocate biomass (assimilates) into the formed reproductive parts of the plant (WNUK et al. [2]). Harvest index is the partitioning of dry matter by plant among biological and economic yield (SHAFI et al. [3]). Also, the harvest index can be defined as been the physiological efficiency and ability of a crop for converting the total dry matter into economic yield (SHARIFI et al. [4]).

Harvest index (HI) can be calculated as the ration of grain yield (Y) and the total above-ground biomass (B) at maturity (HUEHN [5]), than one could estimate the yield (Y) by multiplying the above-ground biomass (B) with harvest index (HI), and finally the above-ground biomass (B) by dividing the yield (Y) with the harvest index (HI):

$$HI(1); Y = B \times HI(2); \quad B(3)$$

These equations are useful for estimation the grain yield for different purposes starting from the above-ground biomass, but also for estimation the above-ground biomass and more precisely the crop residues starting from the obtained yield of grains.

Above-ground biomass represents the crop's accumulated photosynthesis products while harvest index represents the efficiency of the crop to convert photosynthesized products into an economically valuable form (KAWANO [6]).

Among the cereal crops, maize has the highest grain yield potential. In order to explore the high potential of maize plants to produce grain yields it is necessary to know and even more important to understand how the harvest index is influenced by the different environmental factors and management practices.

Drought stress is one of the most important environmental factors in reduction of growth, development and production of plants (KHALILI et al. [7]), this affecting also the harvest index. Among the abiotic factors influencing mostly the maize growth, alongside water stress the soil fertility is of great importance. So, water and soil conditions are two of the most important factors affecting harvest index.

Sustaining maize grain yield increases into the future requires continued reconsideration of current agronomic practices (ROBLES et al. [8]). Among the agronomic practices with a great influence upon the harvest index there are counted the sowing patterns, respectively row spacing and plant density, but also preceding crop and soil tillage.

Planting geometry influences the water use efficiency. Thus, the losses of water through evaporation are minimum in closer row spacing because the leaf canopy is able to cover up the soil surface quickly (SAQIB et al. [9]).

The objective of this paper is to present the results obtained at different hybrids of maize regarding the harvest index under the specific growing conditions from South Romania in the period 2012-2014 and under different sowing conditions, respectively different row spacing, plant densities, preceding crops, and soil tillage.

2. Materials and Methods

Researches performed in 2012. Researches were performed in field experiments at an assortment of ten maize hybrids, respectively: CERA 270 (FAO group 270), CERA 290 (FAO group 290), CERA 370 (FAO group 370), CERA 390 (FAO group 390), CERA 6 (FAO group 390), CERA 420 (FAO group 420), CERA 2504 (FAO group 440), CERA 4505 (FAO group 450), Baragan 48 (FAO group 480), and CERA 10 (FAO group 540).

The maize hybrids were studied in four different locations in South Romania, respectively: Fundulea from Calarasi County, Valcelele from Calarasi County, Visani from Braila County, and Posta Calnau from Buzau County.

From a climatic point of view, the year 2012 could be characterized as warm and dry. The temperatures were higher than multiannual averages, especially in April and August. Regarding the rainfall, except the month May, which was very wet (with rainfall more than double compared to the normal value), the weather until autumn was dry, in July the rainfall being only of 2 mm (about 70 mm deficit).

In average for the ten studied hybrids, the plant density was of 66,000 plants ha⁻¹ in Fundulea location, 62,600 plants ha⁻¹ in Valcelele location, 63,400 plants ha⁻¹ in Visani location, and 62,800 plants ha⁻¹ in Posta Calnau location.

Researches performed in 2013. Researches were performed in field experiments in two locations from South Romania, respectively: Fundulea from Calarasi County, and Moara Domneasca from Ilfov County.

The studied maize hybrids were the following: Cera 440 (FAO group 440), Flanker (FAO group 450), ES Feria (FAO group 550), PR35T36 (FAO group 500), and Cera 540 (FAO group 540). Each hybrid was studied under three row spacing (75 cm, 50 cm, and twin-rows of 75/45 cm) and three plant densities (60,000, 70,000, and 80,000 plants ha⁻¹).

The soil from Fundulea area is chernozem (cambic chernozem soil), while the soil from Moara Domneasca area is reddish preluvosoil.

For the period September 2012 - August 2013, the average temperature was of 12.0°C and the sum of rainfall was of 700.6 mm at Fundulea area, while at Moara Domneasca area the average temperature was of 12.6°C, and the sum of rainfall was of 288.0 mm.

The preceding crop was sunflower. The fertilization was performed with 106 kg of nitrogen ha⁻¹ and 40 kg of phosphorus ha⁻¹. The weed control was performed by the help of herbicides and through one mechanical hoeing.

Researches performed in 2014. Researches performed in the year 2014 were similar to those performed in 2013, in the same locations and under the same experimental and technological conditions.

In 2014, apart from the field experiment continued from 2013, there was also performed a field experiment located at Moara Domneasca area with six maize hybrid studied after two preceding crops (sunflower and maize) and under two soil tillage conditions (plowing and disking). The studied maize hybrids were the following: Mikado (FAO group 550), Korimbos (FAO group 500), Janett (FAO group 550), ES Feria (FAO group 550), ES Metwod (FAO group 380), and ES Antalia (FAO group 450). The fertilization and the weed control were performed as in the case of the other field experiments carried out in 2013 and 2014, respectively the fertilization was performed with 106 kg of nitrogen ha⁻¹ and 40 kg of phosphorus ha⁻¹, and the weed control was performed by the help of herbicides and through one mechanical hoeing.

For the period September 2013 - August 2014, the average temperature was of 10.7°C and the sum of rainfall was of 613.4 mm at Fundulea area, while at Moara Domneasca area the average temperature was of 11.8°C, and the sum of rainfall was of 660.0 mm.

Determination of the harvest index. Harvest index was calculated as the ration of grain yield as dry matter from the total dry matter of the above-ground biomass. The grain yield and the above-ground biomass were determined for individual maize plants. In this respect, eight

plants for each maize hybrid in 2012 and three plants for each experimental variant in 2013 and 2014 were cut at soil level in view to be determined the dry matter. The analysed plants were average and representative plants for each hybrid and experimental variant. The grains were separated from the ear and then dried in the oven for 24 hours at a temperature of 80°C, the same being done for the whole maize plant. The obtained data were statistically processed by analyses of variance.

3. Results and Discussions

Harvest index at different maize hybrids. Under the climatic conditions of the year 2012 in South Romania, the harvest index varied according to hybrid and location, which means different soil and climatic conditions (Figure 1). The highest value of the harvest index was registered at Baragan 48 hybrid, respectively 0.44, while the smallest value of the harvest index was registered at CERA 4505 hybrid, respectively 0.21, both extreme values being statistically significant. There are some maize hybrids with harvest index being very much influenced by growing conditions, more precisely by soil and climatic conditions, with a large variation from one location to another (e.g. hybrids CERA 4505, CERA 207, and Baragan 48). Also, there are some maize hybrids with harvest index being less influenced by growing conditions, with a little variation from one location to another (e.g. hybrids CERA 6, CERA 290, CERA 2504, and CERA 390). In average for the ten studied maize hybrids, the harvest index varied between 0.33 and 0.37 according to location, respectively according to soil and climatic conditions.

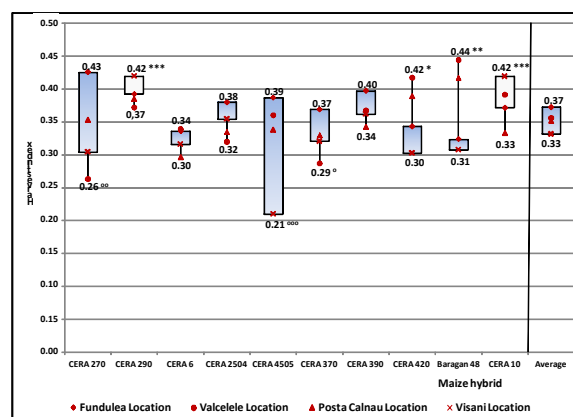


Figure 1. Harvest index at different maize hybrids cultivated in different locations in South Romania in the climatic conditions of 2012

Under favourable soil and climatic conditions registered at Fundulea location, the harvest index registered higher values than those obtained under less favourable soil and climatic conditions registered at Moara Domneasca location (Figure 2). Both in 2013 and 2014 climatic conditions and in both experimental locations (Fundulea and Moara Domneasca), the harvest index registered the highest values at PR35T36 and ES Feria hybrids, while the smallest value was registered at Flanker hybrid. Under the more favourable climatic conditions of the year 2014, the harvest index registered larger variations according to location compared to those values obtained under the less favourable climatic conditions of the year 2013. Thus, in average for the five studied hybrids of maize, the harvest index varied between 0.36 and 0.38 in the year 2013 (Figure 2.a), while in the year 2014 the harvest index varied between 0.36 and 0.41 (Figure 2.b).

The hybrid with the lowest variation of the harvest index and the most stable one in time and space (in both experimental years and in both experimental locations) was ES Feria, with variation of the harvest index from 0.39 to 0.40 in 2013 and from 0.41 to 0.42 in 2014. The hybrid with the largest variation of the harvest index in time and space was PR35T36, with variation of the harvest index from 0.39 to 0.40 in 2013 and from 0.37 to 0.49 in 2014.

The findings according to what the maize hybrids have significant variation of the harvest index are similar to those obtained by different authors (SIKANDAR et al. [11]; SHARIFI et al. [4]; SAQIB et al [9]).

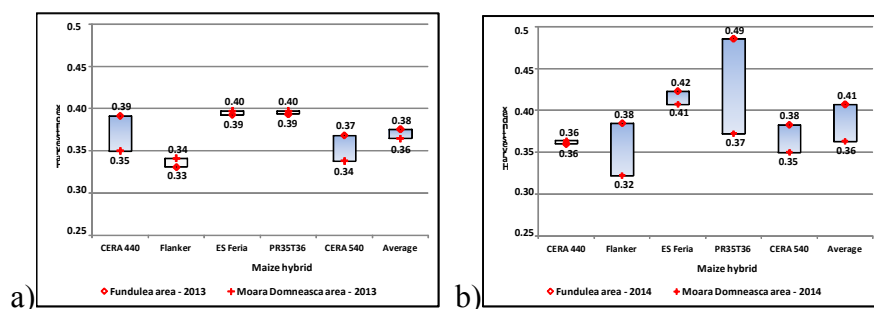


Figure 2. Harvest index at different maize hybrids cultivated in different locations in South Romania in the climatic conditions of 2013 (a) and 2014 (b)

Harvest index at different row spacing and plant density. At Fundulea location and in the year 2013, the highest value of the harvest index (0.44) was registered at row spacing of 75 cm and at plant density of 60.000 plants ha⁻¹, while the smallest values of the harvest index (0.34) were registered at narrow rows, respectively at 50 cm and twin rows of 75/45 cm, and at higher plant density, respectively at 70.000 and 80.000 plants ha⁻¹ (Figure 3.a). The highest average values of the harvest index were registered at row spacing of 75 cm (0.41) and at plant density of 60.000 plants ha⁻¹ (0.39).

In the year 2014 and at Fundulea location, the better climatic conditions determined a wider variation of the harvest index according to plant density, the values being higher than those registered in 2013 (Figure 3.b). The highest value of the harvest index (0.48) was registered at row spacing of 75 cm and at plant density of 70.000 plants ha⁻¹. The smallest value of the harvest index (0.35) was registered at twin-rows of 75/45 cm and at plant density of 80.000 plants ha⁻¹. The highest average values of harvest index were registered at row spacing of 50 cm (0.42) and at plant density of 70.000 plants ha⁻¹ (0.44).

At Fundulea location, the smallest variation of the harvest index according to plant density were registered at narrow rows in 2013, respectively at row spacing of 50 cm and twin-rows of 75/45 cm, and at row spacing of 50 cm in 2014.

At Fundulea location, the average value of the harvest index in 2013 was of 0.38, while in 2014 this was of 0.41, which means that in 2014 the harvest index was higher, but with larger variations according to plant density.

At Moara Domneasca location and in the year 2013, the highest value of the harvest index (0.40) was registered at narrow rows, respectively at row spacing of 50 cm and plant density of 80.000 plants ha⁻¹, and at twin-rows of 75/45 cm and plant density of 60.000 plants ha⁻¹ (Figure 4.a). The smallest values of the harvest index (0.31) were registered at row spacing of 75 cm and plant density of 60.000 plants ha⁻¹. The highest average values of the harvest index were registered at narrow rows (0.39 for twin-rows of 75/45 cm and 0.38 for row spacing of 50 cm) and at plant density of 80.000 plants ha⁻¹ (0.37).

In the year 2014 and at Moara Domneasca location, the better climatic conditions caused a lower variation of the harvest index, but the values were higher than those registered in 2013 (Figure 4.b). The highest value of the harvest index (0.40) was registered at row spacing of 50 cm and plant density of 70.000 plants ha⁻¹. The smallest value of the harvest index (0.33) was registered at twin-rows of 75/45 cm and plant density of 80.000 plants ha⁻¹. The highest average values of the harvest index were registered at row spacing of 50 cm (0.39) and at plant density of 60.000 and 70.000 plants ha⁻¹ (0.37).

At Moara Domneasca location, the smallest variations of the harvest index according to plant density were registered in 2013 at twin-rows of 75/45 cm and in 2014 at narrow rows (row spacing of 50 cm and twin-rows of 75/45 cm).

The average value of the harvest index was of 0.36 in 2013, while in 2014 this was of 0.37, which means that in 2014 the harvest index was higher, but with lower variations according to plant density.

The average values of the harvest index in both locations and in both experimental years registered the smallest variations at narrow rows, especially at row spacing of 50 cm (Figure 5.a). The largest variations were registered at row spacing of 75 cm, here being registered also the smallest and the highest average values of the harvest index.

Regarding the average values of the harvest index at different plant densities, the smallest variations were registered at plant population of 80.000 plants ha⁻¹ (Figure 5.b). The highest average values of the harvest index were registered at plant density of 70.000 plants ha⁻¹, while the smallest average values of the harvest index were registered at plant density of 60.000 plants ha⁻¹. These data are comparable to those obtained by SHAFI et al. (2012), who obtained the highest harvest index in the treatment of 65.000 plants ha⁻¹ and the lowest in 45.000 plants ha⁻¹.

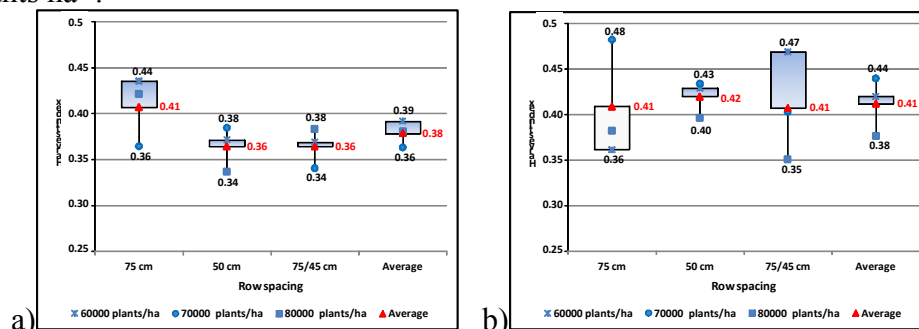


Figure 3. Harvest index at maize under different row spacing and plant density registered at Fundulea in the years 2013 (a) and 2014 (b) (average data for five maize hybrids)

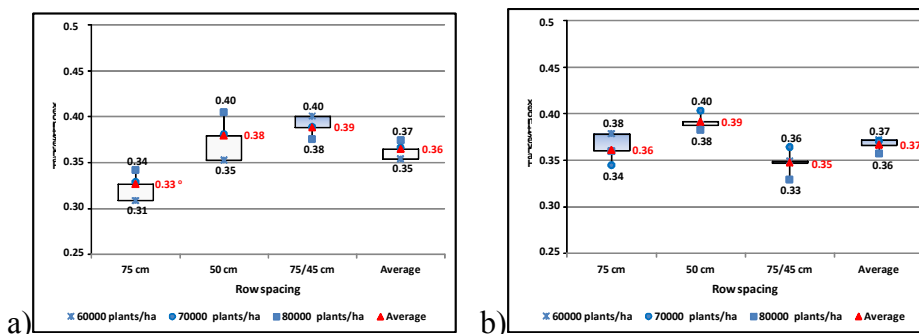


Figure 4. Harvest index at maize under different row spacing and plant density registered at Moara Domneasca in the years 2013 (a) and 2014 (b) (average data for five maize hybrids)

Harvest index at maize in different growing conditions

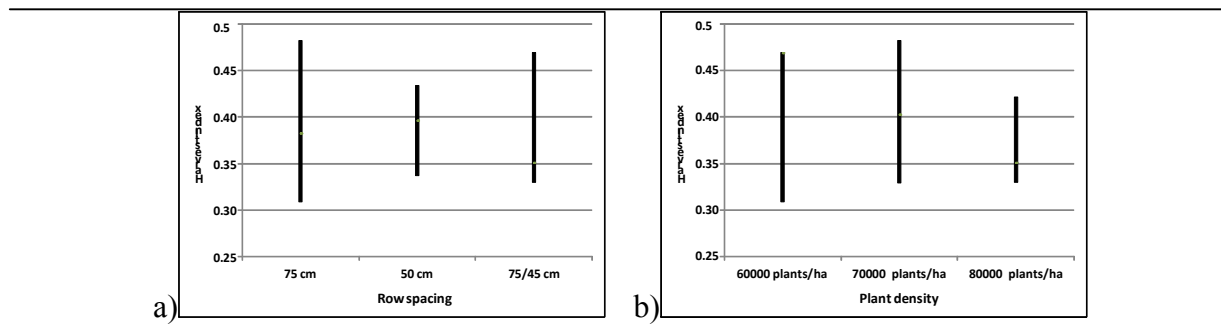


Figure 5. The average values of the harvest index at maize under different row spacing (a) and plant density (b), registered at Fundulea and Moara Domneasca in the years 2013 and 2014

Harvest index at different preceding crops and soil tillage. Compared to sunflower as preceding crop, maize as preceding crop determined higher values of the harvest index but also a larger variation of the harvest index, according to the cultivated maize hybrid (Figure 6). As concerning the soil tillage, compared to plowing the disking determined higher values of the harvest index but also a larger variation of the harvest index, according to the cultivated maize hybrid.

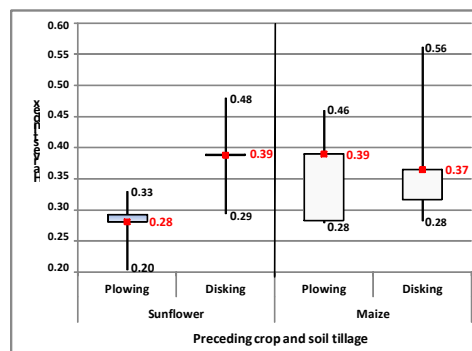


Figure 6. Harvest index at maize after different preceding crops and under different soil tillage registered at Moara Domneasca in the year 2014 (average data for six maize hybrids)

Correlation of harvest index with plant height. The harvest index correlates differently to plant height according to locations where the hybrids were cultivated, respectively to soil and climatic conditions. In some areas, the harvest index correlates negatively to plant height (Figure 7.a and Figure 7.c), respectively when the climatic conditions were less favourable for growing maize. On the contrary, in the area with better climatic conditions associated also with good soil conditions, the harvest index tends to correlates positively with plant height up to a certain level (Figure 7.b).

These correlations sustain the idea that shorter plants could have higher harvest index, which means a higher grain yielding capacity, respectively the maize plant use more assimilates for grain yielding instead to build the large stems.

Regardless of growing conditions, the coefficient of determination (R^2) has small values. However the values of the R^2 are higher under better growing conditions.

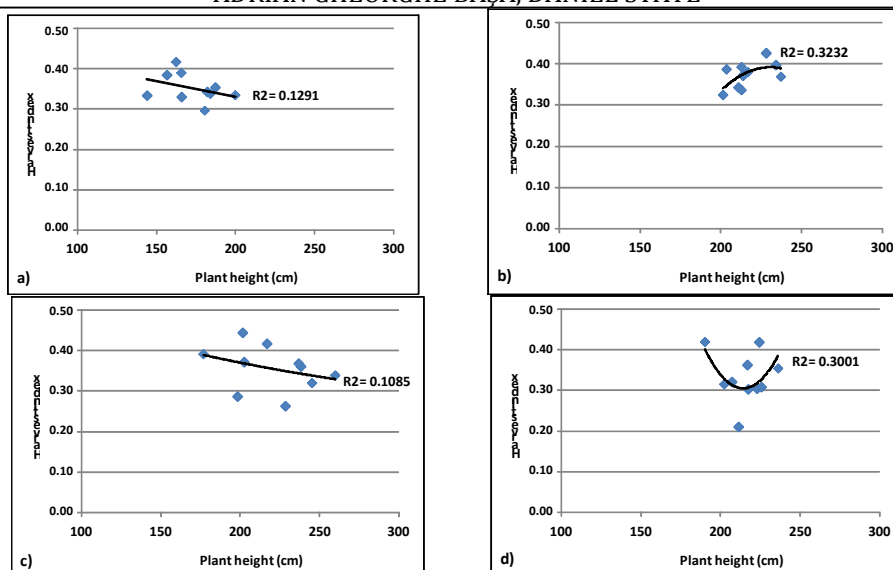


Figure 7. Harvest index at maize in function of plant height, in the year 2012 and in different locations: a) Posta Calnau location; b) Fundulea area; c) Valcelele area; d) Visani area (average data for ten maize hybrids)

Correlation of harvest index with stem diameter at soil level. As in the case of plant height, the harvest index correlates differently to stem diameter at soil level according to locations where the hybrids were cultivated, respectively to soil and climatic conditions (Figure 8).

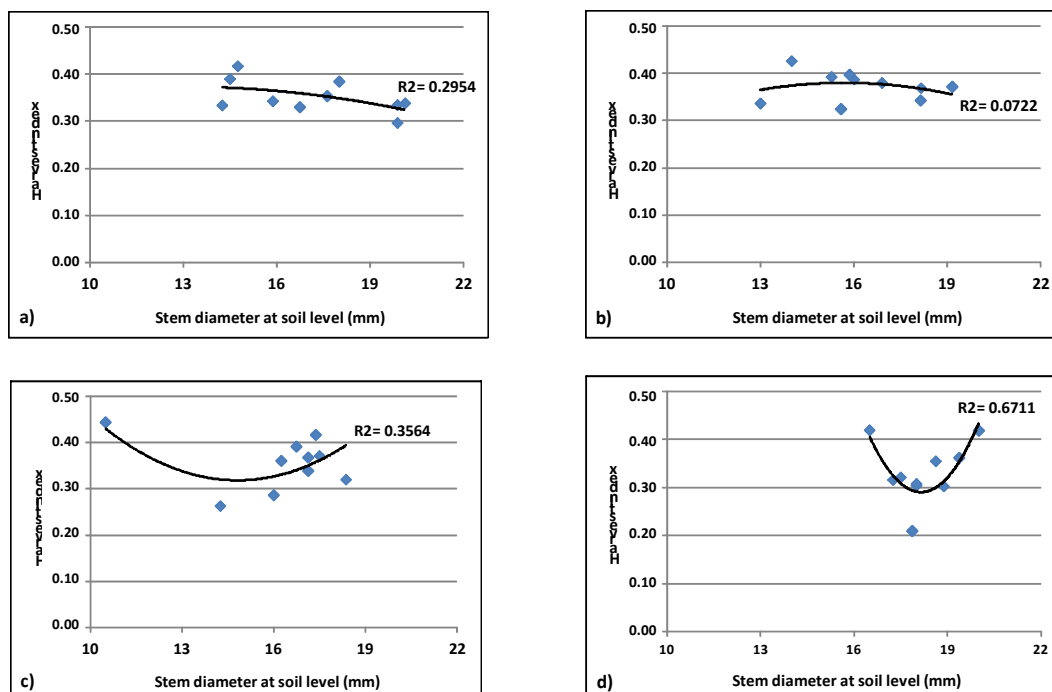


Figure 8. Harvest index at maize in function of stem diameter at soil level, in the year 2012 and in different locations: a) Posta Calnau location; b) Fundulea area; c) Valcelele area; d) Visani area (average data for ten maize hybrids)

4. Conclusions

In our field experiments, the average values of the harvest index at maize varied between 0.20 to 0.56, more frequently between 0.3 and 0.5. The harvest index registered different values according to maize hybrid, climatic and soil conditions, row spacing, plant density, preceding crop and soil tillage. Harvest index at maize registered higher values under more favourable soil and climatic conditions than under less favourable conditions. When the climatic conditions became more favorable, the harvest index registered larger variations according to soil conditions; under less favorable climatic conditions the harvest index registered fewer variations according to soil conditions. Also, better climatic conditions determined a larger variation of the harvest index according to plant density and row spacing. There are maize hybrids with harvest index being less influenced by growing conditions, which is relatively stable from one year to another. There are also maize hybrids with harvest index much influenced by growing conditions, with high values of the harvest index when the growing conditions are favourable and with small values of the harvest index when the growing conditions are less favourable. Narrow rows, especially the row spacing of 50 cm, compared to the row spacing of 75 cm tended to determine smaller variation of the harvest index according to plant density and maize hybrid. The highest average values of the harvest index were registered at plant density of 70.000 plants ha⁻¹, while the smallest average values of the harvest index were registered at plant density of 60.000 plants ha⁻¹. Maize as preceding crop determined higher values of the harvest index than sunflower, but also a larger variation of the harvest index, according to the cultivated maize hybrid. Disking determined higher values of the harvest index than plowing, but also a larger variation of the harvest index, according to the cultivated maize hybrid. Harvest index correlated differently to plant height and to stem diameter at soil level according to locations where the hybrids were cultivated, respectively to soil and climatic conditions. Harvest index tended to correlate negatively to plant height when the climatic conditions were less favourable for growing maize and tended to correlate positively up to a certain level when the climatic conditions were more favourable, especially when these were associated with favourable soil conditions.

5. Acknowledgements

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